

For multivariate and an software idea

Yicheng Xu

16th June 2026

1 Differential Parameter LMI

Consider the general Differential Parameter LMI (DP-LMI):

$$\sum_{i=1}^N R_i(\rho) \sum_{j=1}^{\ell} \frac{\partial x_i(\rho)}{\partial \rho_j} \dot{\rho}_j + F_0(\rho) + \sum_{i=1}^N F_i(\rho) x_i(\rho) \prec 0, \quad (1)$$

where $\rho \in \mathbb{R}^{\ell}$ is the parameter, and $x_i(\rho) \in \mathbb{R}$ are the decision variables. For example, the most simple case is the following:

Example 1: For a continuous-time LPV system $\dot{x}(t) = A(\rho)x(t)$, the stability condition is given by the following DP-LMI:

$$\begin{bmatrix} \sum_{j=1}^{\ell} \dot{\rho}_j \frac{\partial P(\rho)}{\partial \rho_j} + A^T(\rho)P(\rho) + P(\rho)A(\rho) & 0 \\ 0 & -P(\rho) \end{bmatrix} \prec 0. \quad (2)$$

Here $R_i(\rho) \equiv I$.

Typically, the DP-LMI is considered on a continuous set:

Assumption 1: The parameter ρ and its derivative $\dot{\rho}$ belongs to the set:

$$(\rho \times \dot{\rho}) \in \Theta \times \Phi.$$

1.1 Boundness on the $\dot{\rho}$

In reality, the parameter ρ is often represented by a physical quantity, which normally is bounded. However, the derivative $\dot{\rho}$ is not necessarily bounded. Therefore, to make the problem more practical, we can consider for a scalar case, i.e., $\ell = 1$:

1. Bounded $\dot{\rho}$: $\dot{\rho} \in [\underline{\nu}, \bar{\nu}]$.

(1) can then be solved on the vertices of the set Φ :

$$\begin{bmatrix} \underline{\nu} \sum_{i=1}^N R_i(\rho) \sum_{j=1}^{\ell} \frac{\partial x_i(\rho)}{\partial \rho_j} + F_0(\rho) + \sum_{j=1}^N F_j(\rho) x_j(\rho) & 0 \\ 0 & \bar{\nu} \sum_{i=1}^N R_i(\rho) \sum_{j=1}^{\ell} \frac{\partial x_i(\rho)}{\partial \rho_j} + F_0(\rho) + \sum_{j=1}^N F_j(\rho) x_j(\rho) \end{bmatrix} \prec 0.$$

2. One-side bounded $\dot{\rho}$: $\dot{\rho} \in [-\infty, \bar{\nu}]$. This case can appear, for example, in saturation [1]. It is sufficient to instead, consider the following DP-LMI $\forall \rho \in \Theta$:

$$\begin{bmatrix} \bar{\nu} \sum_{i=1}^N R_i(\rho) \sum_{j=1}^{\ell} \frac{\partial x_i(\rho)}{\partial \rho_j} + F_0(\rho) + \sum_{j=1}^N F_j(\rho) x_j(\rho) & 0 \\ 0 & \sum_{i=1}^N R_i(\rho) \sum_{j=1}^{\ell} \frac{\partial x_i(\rho)}{\partial \rho_j} \end{bmatrix} \prec 0,$$

The other way around $\dot{\rho} \in [\underline{\nu}, +\infty]$ follows the same way, but flips the sign of the second inequality

$$\sum_{i=1}^N R_i(\rho) \sum_{j=1}^{\ell} \frac{\partial x_i(\rho)}{\partial \rho_j} \succeq 0.$$

3. Unbounded $\dot{\rho}$: $\dot{\rho} \in (-\infty, +\infty)$. This case can be considered, for example, in arbitrary switching system [2], or uncertain systems [3]. Then it has no choice but to consider the following DP-LMI $\forall \rho \in \Theta$:

$$F_0(\rho) + \sum_{j=1}^N F_i(\rho)x_i(\rho) \prec 0,$$

$$\sum_{i=1}^N R_i(\rho) \sum_{j=1}^{\ell} \frac{\partial x_i(\rho)}{\partial \rho_j} = 0.$$

There are some workarounds that might be useful to decouple the relation of $R(\rho)$ and $F_i(\rho)$, for example, by applying projection lemma to (2):

$$\begin{bmatrix} 0 & P(\rho) \\ P(\rho) & 0 \end{bmatrix} + \begin{bmatrix} I \\ I \end{bmatrix} E(\rho) \begin{bmatrix} A(\rho) & -I \end{bmatrix} \prec 0,$$

$$\frac{\partial E(\rho)}{\partial \rho_i} = 0.$$

But it is limited to case-wise analysis, and generally speaking, it is hard to consider the unbounded $\dot{\rho}$ case.

In summary, in both bounded case and one-side bounded case, the DP-LMI can be solved by considering the vertices of the set Φ , and therefore, augmenting it to a larger DP-LMI into a diagonal form. It is then sufficient to only consider the following DP-LMI

$$Q(x, \rho) := \sum_{i=1}^N R_i(\rho) \sum_{j=1}^{\ell} \frac{\partial x_i(\rho)}{\partial \rho_j} + F_0(\rho) + \sum_{i=1}^N F_i(\rho)x_i(\rho) \prec 0, \quad (3)$$

Note that the $\dot{\rho}$ disappears in (3), because it is considered on the vertices of the set Φ , and therefore becomes part of the coefficients R_i . But the decision variable $R(\rho)$ is still a function of ρ , which needs to be solved for all $\rho \in \Theta$.

1.2 On solving (3) on $\rho \in \Theta$

Inspect the case the parameter ρ is scalar, i.e., $\ell = 1$. In (3), x is the decision variable, and $\rho \in \Theta$ is the parameter. The problem is to find a solution $x(\rho)$ for all $\rho \in \Theta$.

Theorem 1: Suppose Assumption 1 holds and $\ell = 1$, Θ is compact, and $R_i(\rho), F_0(\rho), F_i(\rho)$ are *Lipschitz continuous* function in ρ , then the following three statements are equivalent:

1. There exists *Lipschitz continuously differentiable* solution $x(\rho) \in C^{1,1}$, i.e., $\frac{\partial x(\rho)}{\partial \rho}$ Lipschitz, with

$$l_0 := \max_{\rho \in \Theta} \left\| \frac{\partial x(\rho)}{\partial \rho} \right\|, l_1 := \text{Lip}_{\rho \in \Theta} \left\| \frac{\partial x(\rho)}{\partial \rho} \right\|$$

to (3) for all $\rho \in \Theta$.

2. Let the Lipschitz constant of the residue linear spline approximation of $F_0(\rho), F_i(\rho), R_i(\rho)$:

$$r_0^k = \text{Lip}_{\rho \in [\rho^k, \rho^{k+1}]} \|F_0(\rho) - F_0^L(\rho)\|,$$

$$r_1^k = \text{Lip}_{\rho \in [\rho^k, \rho^{k+1}]} \sqrt{\sum_{i=1}^N \|F_i(\rho) - F_i^L(\rho)\|^2},$$

$$r_2^k = \text{Lip}_{\rho \in [\rho^k, \rho^{k+1}]} \sqrt{\sum_{i=1}^N \|R_i(\rho) - R_i^L(\rho)\|^2},$$

There exists a division of $\Theta : \mathcal{D} = \{\rho^1, \dots, \rho^{K+1}\}$ such that the following LMIs hold for all $k = 1, \dots, K$:

$$\begin{bmatrix} -d^k I_N & x(\rho^k) \\ * & -d^k \end{bmatrix} \preceq 0, \quad \forall k = 1, \dots, K+1, \quad (4a)$$

$$Q_1^k \prec -(r_0^k + r_1^k d^k + r_2^k l_0) \Delta^k I_n, \quad \forall k = 1, \dots, K, \quad (4b)$$

$$Q_3^k \prec -(r_0^k + r_1^k d^{k+1} + r_2^k l_0) \Delta^k I_n, \quad \forall k = 1, \dots, K, \quad (4c)$$

$$Q_2^k \prec -[2r_0^k + r_1^k (d^k + d^{k+1}) + 2r_2^k l_0] \Delta^k I_n, \quad \forall k = 1, \dots, K, \quad (4d)$$

where $\Delta^k = \rho^{k+1} - \rho^k$, and

$$\begin{aligned} Q_1^k &= \sum_{i=1}^N R_i(\rho^k) \frac{x_i(\rho^{k+1}) - x_i(\rho^k)}{\rho^{k+1} - \rho^k} + F_0(\rho^k) + \sum_{i=1}^N F_i(\rho^k) x_i(\rho^k), \\ Q_3^k &= \sum_{i=1}^N R_i(\rho^{k+1}) \frac{x_i(\rho^{k+1}) - x_i(\rho^k)}{\rho^{k+1} - \rho^k} + F_0(\rho^{k+1}) + \sum_{i=1}^N F_i(\rho^{k+1}) x_i(\rho^{k+1}), \\ Q_2^k &= \sum_{i=1}^N [R_i(\rho^k) + R_i(\rho^{k+1})] \frac{x_i(\rho^{k+1}) - x_i(\rho^k)}{\rho^{k+1} - \rho^k} + F_0(\rho^k) + F_0(\rho^{k+1}) + \sum_{i=1}^N F_i(\rho^{k+1}) x_i(\rho^k) + F_i(\rho^k) x_i(\rho^{k+1}). \end{aligned}$$

3. There exists an upper bound $\bar{\Delta}$ such that for any division $\mathcal{D}$ of Θ with $\max_{k=1, \dots, K} \Delta^k \leq \bar{\Delta}$, (4) holds for all $k = 1, \dots, K$ with a stronger requirement:

$$d^k \rightarrow d, r_0^k \rightarrow r_0 = 2 \operatorname{Lip}_{\rho \in \Theta} \|F_0(\rho)\|, \quad (5a)$$

$$r_1^k \rightarrow r_1 = 2 \operatorname{Lip}_{\rho \in \Theta} \sqrt{\sum_{i=1}^N \|F_i(\rho)\|^2}, \quad (5b)$$

$$r_2^k \rightarrow r_2 = 2 \operatorname{Lip}_{\rho \in \Theta} \|R_i(\rho)\|. \quad (5c)$$

Proof: The proof follows the logic $3) \Rightarrow 2) \Rightarrow 1) \Rightarrow 3)$.

- $3) \Rightarrow 2)$: This is trivial, because the condition in 3) is stronger than that in 2).
- $2) \Rightarrow 1)$: We only need to focus on one sub-interval $[\rho^k, \rho^{k+1}]$, and the rest follows by substituting $k = 1, \dots, K$. Denote $\alpha := \frac{\rho^{k+1} - \rho}{\rho^{k+1} - \rho^k}$, then the convexity of the LMI leads to:

$$\begin{aligned} \alpha^2 Q_1^k + (1 - \alpha) Q_2^k + (1 - \alpha)^2 Q_3^k &\prec - [r_0^k + r_1^k (\alpha d^k + (1 - \alpha) d^{k+1}) + r_2^k l] \Delta^k I_n, \\ \sum_{i=1}^N R_i^L(\rho) \frac{x(\rho^{k+1}) - x(\rho^k)}{\rho^{k+1} - \rho^k} + F_0^L(\rho) + \sum_{i=1}^N F_i^L(\rho) x_i^L(\rho) &\prec - [r_0^k + r_1^k (\alpha d^k + (1 - \alpha) d^{k+1}) + r_2^k l] \Delta^k I_n, \end{aligned} \quad (6)$$

(4a) implies that $\|x(\rho^k)\| \leq d^k$, of which the subadditivity leads to:

$$\|x^L(\rho)\| = \|\alpha x(\rho^k) + (1 - \alpha) x(\rho^{k+1})\| \leq \alpha \|x(\rho^k)\| + (1 - \alpha) \|x(\rho^{k+1})\| \leq \alpha d^k + (1 - \alpha) d^{k+1}.$$

Then the Cauchy-Schwarz inequality applied to $\sum_{i=1}^N [F_i(\rho) - F_i^L(\rho)] x_i^L(\rho)$ leads to:

$$\left\| \sum_{i=1}^N [F_i(\rho) - F_i^L(\rho)] x_i^L(\rho) \right\| \leq \sqrt{\sum_{i=1}^N \|F_i(\rho) - F_i^L(\rho)\|^2} \|x^L(\rho)\| \leq r_1^k \Delta^k (\alpha d^k + (1 - \alpha) d^{k+1}). \quad (7)$$

Similarly, applying the Cauchy-Schwarz inequality to $\sum_{i=1}^N [R_i(\rho) - R_i^L(\rho)] \frac{x_i(\rho^{k+1}) - x_i(\rho^k)}{\rho^{k+1} - \rho^k}$ leads to:

$$\left\| \sum_{i=1}^N [R_i(\rho) - R_i^L(\rho)] \frac{x_i(\rho^{k+1}) - x_i(\rho^k)}{\rho^{k+1} - \rho^k} \right\| \leq \sqrt{\sum_{i=1}^N \|R_i(\rho) - R_i^L(\rho)\|^2} \left\| \frac{x(\rho^{k+1}) - x(\rho^k)}{\rho^{k+1} - \rho^k} \right\| \leq r_2^k \Delta^k l_0. \quad (8)$$

The last step is to consider the term $F_0(\rho) - F_0^L(\rho)$:

$$\|F_0(\rho) - F_0^L(\rho)\| \leq r_0^k \Delta^k. \quad (9)$$

Considering (6) to (9), we have $Q(x^L, \rho) \prec 0$ holds on $[\rho^k, \rho^{k+1}]$.

- $1) \Rightarrow 3)$:
We rely on the Extreme Value theorem. Given that $R_i(\rho), F_0(\rho), F_i(\rho), x(\rho)$ Lipschitz continuous, the following statements hold:

i) $Q(x, \rho)$ has upper bound on compact Θ :

$$\begin{aligned} & Q(x, \rho) \text{ Lipschitz continuous on } \Theta \\ & \Rightarrow \|Q(x, \rho)\| \text{ Continuous on } \Theta \\ & \xrightarrow[\text{theorem}]{\text{Extreme value}} \exists a > 0, Q(x, \rho) \preceq -aI, \forall \rho \in \Theta. \end{aligned} \quad (10)$$

ii) $\|x(\rho)\|$ is bounded by some $d < \infty$ on Θ , using Extreme value theorem.

iii) $\sqrt{\sum_{i=1}^N \|R_i(\rho)\|^2}$ is bounded by some $D < \infty$ on Θ , using Extreme value theorem.

Take the values in (5). It is sufficient to demonstrate that such

$$\bar{\Delta} := \frac{a}{r_0 + 2r_1d + r_2l_0 + 2Dl_1}$$

satisfies the requirement in 3). Evaluate $Q(x, \rho) \preceq -aI$ at $\rho = \rho^k$ gives:

$$\begin{aligned} & \sum_{i=1}^N R_i(\rho^k) \frac{\partial x_i(\rho^k)}{\partial \rho} + F_0(\rho^k) + \sum_{i=1}^N F_i(\rho^k) x_i(\rho^k) \preceq -aI \prec -(r_0 + 2r_1d + r_2l_0 + 2Dl_1) \Delta^k I, \\ & \sum_{i=1}^N R_i(\rho^k) \frac{x_i(\rho^{k+1}) - x_i(\rho^k)}{\rho^{k+1} - \rho^k} + F_0(\rho^k) + \sum_{i=1}^N F_i(\rho^k) x_i(\rho^k) \prec -(r_0 + 2r_1d + r_2l_0) \Delta^k I \preceq -(r_0 + r_1d + r_2l_0) \Delta^k I. \end{aligned} \quad (11)$$

which becomes (4b). Similar argument applies to (4c) where $Q(x, \rho)$ is evaluated at $\rho = \rho^{k+1}$

$$\sum_{i=1}^N R_i(\rho^{k+1}) \frac{x_i(\rho^{k+1}) - x_i(\rho^k)}{\rho^{k+1} - \rho^k} + F_0(\rho^{k+1}) + \sum_{i=1}^N F_i(\rho^{k+1}) x_i(\rho^{k+1}) \prec -(r_0 + 2r_1d + r_2l_0) \Delta^k I \preceq -(r_0 + r_1d + r_2l_0) \Delta^k I. \quad (12)$$

The only thing left is to show (4d) holds, which can be written as the sum of (11) and (12) plus the residue:

$$\begin{aligned} Q_2^k + \sum_{i=1}^N [F_i(\rho^{k+1}) - F_i(\rho^k)] x_i(\rho^k) + \sum_{i=1}^N [F_i(\rho^k) - F_i(\rho^{k+1})] x_i(\rho^{k+1}) & \prec -(2r_0 + 4r_1d + 2r_2l_0) \Delta^k I, \\ Q_2^k & \prec -(2r_0 + 2r_1d + 2r_2l_0) \Delta^k I. \end{aligned}$$

■

Remark 1: Statement 1) is the original problem, and statement 3) states that the original problem can be solved by gridding technique with a sufficiently fine grid. For practical purposes, statement 2) is proposed as a numerical method to solve the original problem, which is less conservative than statement 3). For example, when the system matrices F_0, F_i, R_i are already linear spline in ρ , the residue terms $r_0^k = 0, r_1^k = 0, r_2^k = 0, \forall k = 1, \dots, K$, and therefore, (4b) to (4d) becomes $Q_1^k \prec 0, Q_3^k \prec 0, Q_2^k \prec 0$, which is much more less conservative.

Given a system matrices $F_0(\rho), F_i(\rho), R_i(\rho)$, and a division $\mathcal{D}$ of Θ , the Lipschitz constants r_0^k, r_1^k, r_2^k can be computed by numerical differentiation. Even though it is upper bounded by r_0, r_1, r_2 defined in 3), in practice, a well-chosen division $\mathcal{D}$ can lead to much smaller r_0^k, r_1^k, r_2^k , and therefore, less conservative results.

For only parameter-dependent problem, where $R_i(\rho) \equiv 0$, the condition in statement 1) can be further relaxed only requiring x be lipschitz continuous and statement 2) remains the same with $r_2^k \equiv 0$.

1.3 Extension to the multivariate case and the useful lemma

Theorem 1 only applies to the scalar parameter case, i.e., $\ell = 1$. For the multivariate case, i.e., $\ell > 1$, the equivalence still holds, but the condition in statement 2) becomes more complicated. The multivariate interpolation of a multivariate function is defined as:

$$P(\rho_1, \dots, \rho_\ell) = \frac{1}{\prod_{i=1}^\ell (\rho_i^{k+1} - \rho_i^k)} \sum_{\nu \in \{0,1\}^\ell} \left[P(\rho_1^{k+\nu_1}, \dots, \rho_\ell^{k+\nu_\ell}) \prod_{j|\nu_j=1} (\rho_j - \rho_j^k) \prod_{j|\nu_j=0} (\rho_j^{k+1} - \rho_j) \right].$$

The first-order partial derivative with respect to ρ_i is:

$$\frac{\partial P}{\partial \rho_i} = \frac{1}{\prod_{j=1}^{\ell} (\rho_j^{k+1} - \rho_j^k)} \sum_{\nu \in \{0,1\}^{\ell-1}} \left[P(\rho_1^{k+\nu_1}, \dots, \rho_i^{k+1}, \dots, \rho_{\ell}^{k+\nu_{\ell-1}}) \prod_{j|\nu_j=1} (\rho_j - \rho_j^k) \prod_{j|\nu_j=0} (\rho_j^{k+1} - \rho_j) \right. \\ \left. - P(\rho_1^{k+\nu_1}, \dots, \rho_i^k, \dots, \rho_{\ell}^{k+\nu_{\ell-1}}) \prod_{j|\nu_j=1} (\rho_j - \rho_j^k) \prod_{j|\nu_j=0} (\rho_j^{k+1} - \rho_j) \right].$$

Instead of considering the convex combination of three LMIs, we need to consider the convex combination of 3^ℓ LMIs, and replace Δ^k with the measurement of the sub-interval in multivariate domain, which is: $\Delta^{k_1, k_2, \dots, k_\ell} = \prod_{i=1}^{\ell} (\rho_i^{k_i+1} - \rho_i^{k_i})$.

Revisit the useful lemma in CDC 2024 [4]:

Lemma 1: If there exists $S \in \mathbb{R}^{n \times n}$ such that:

$$\begin{bmatrix} L_1 & S^T \\ S & L_3 \end{bmatrix} \prec 0, \quad (13a)$$

$$L_2 \preceq S^T + S, \quad (13b)$$

then the following LMI holds for $\alpha \in [0, 1]$

$$\alpha^2 L_1 + (1 - \alpha) L_2 + (1 - \alpha)^2 L_3 \prec 0. \quad (14)$$

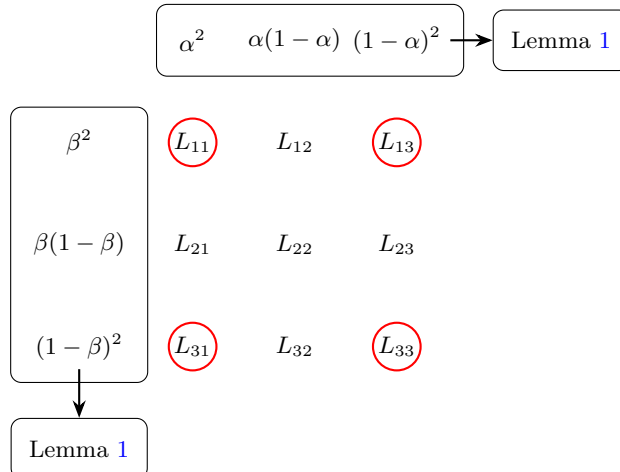
Of course, statement 2) in Theorem 1 can be further relaxed using Lemma 1, which replace (4b) to (4d) by the following LMIs:

$$\begin{bmatrix} Q_1^k & (S^k)^T \\ * & Q_3^k \end{bmatrix} \prec \begin{bmatrix} -(r_0^k + r_1^k d^k + r_2^k l_0) \Delta^k I_n & 0 \\ 0 & -(r_0^k + r_1^k d^{k+1} + r_2^k l_0) \Delta^k I_n \end{bmatrix}, \quad \forall k = 1, \dots, K, \\ Q_2^k \preceq (S^k)^T + S^k - [2r_0^k + r_1^k (d^k + d^{k+1}) + 2r_2^k l_0] \Delta^k I_n, \quad \forall k = 1, \dots, K.$$

Lemma 1 reduces the conservatism of the gridding technique especially when the gridding is coarse. However, it seems hard to find a similar lemma for the case of multivariate ρ , i.e., $\ell > 1$. Let's take a look at the **two** parameter case, i.e., $\ell = 2$.

$$\begin{aligned} & [\alpha\beta A_{11} + \alpha(1-\beta)A_{12} + (1-\alpha)\beta A_{21} + (1-\alpha)(1-\beta)A_{22}] [\alpha\beta B_{11} + \alpha(1-\beta)B_{12} + (1-\alpha)\beta B_{21} + (1-\alpha)(1-\beta)B_{22}] \\ &= \alpha^2\beta^2 A_{11}B_{11} + \alpha^2(1-\beta)\beta [A_{11}B_{12} + A_{12}B_{11}] + \alpha(1-\alpha)\beta^2 [A_{11}B_{21} + A_{21}B_{11}] + (1-\alpha)^2(1-\beta)^2 A_{22}B_{22} \\ &+ \alpha(1-\alpha)\beta(1-\beta) [A_{11}B_{22} + A_{12}B_{21} + A_{21}B_{12} + A_{22}B_{11}] + \alpha^2(1-\beta)^2 A_{12}B_{12} \\ &+ \alpha(1-\alpha)(1-\beta)^2 [A_{12}B_{22} + A_{22}B_{12}] + (1-\alpha)^2\beta^2 A_{21}B_{21} + (1-\alpha)^2\beta(1-\beta) [A_{21}B_{22} + A_{22}B_{21}] \\ &= \begin{matrix} \beta^2\alpha^2 L_{11} & +\beta^2\alpha(1-\alpha)L_{12} & +\beta^2(1-\alpha)^2 L_{13} \\ +\beta(1-\beta)\alpha^2 L_{21} & +\beta(1-\beta)\alpha(1-\alpha)L_{22} & +\beta(1-\beta)(1-\alpha)^2 L_{23} \\ +(1-\beta)^2\alpha^2 L_{31} & +(1-\beta)^2\alpha(1-\alpha)L_{32} & +(1-\beta)^2(1-\alpha)^2 L_{33} \end{matrix} \end{aligned}$$

The gridding technique will consider the following (3×3) 9 points.



Note that for sure at least four points these LMIs are required to be negative definite:

$$L_{11}, L_{13}, L_{31}, L_{33} \prec 0,$$

as they are the vertices of the set $[0, 1]^2$. However, the other five points are not necessarily required to be negative definite. Without knowing the exact structure of L_{ij} , from symmtry, it is reasonable to expect a condition that is symmetric in $L_{12}, L_{21}, L_{22}, L_{23}, L_{32}$.

Lemma 2: Suppose there exists $S_{ij} \in \mathbb{R}^{n \times n}, \forall i, j = 1, 2, 3$ such that:

$$\begin{bmatrix} L_{11} & S_1^T \\ S_1 & L_{13} \end{bmatrix} \prec 0, \quad 3L_{12} \preceq S_1^T + S_1, \quad (15a)$$

$$\begin{bmatrix} L_{31} & S_2^T \\ S_2 & L_{33} \end{bmatrix} \prec 0, \quad 3L_{32} \preceq S_2^T + S_2, \quad (15b)$$

$$\begin{bmatrix} L_{11} & S_3^T \\ S_3 & L_{31} \end{bmatrix} \prec 0, \quad 3L_{21} \preceq S_3^T + S_3, \quad (15c)$$

$$\begin{bmatrix} L_{13} & S_4^T \\ S_4 & L_{33} \end{bmatrix} \prec 0, \quad 3L_{23} \preceq S_4^T + S_4, \quad (15d)$$

$$\begin{bmatrix} L_{11} & S_5^T \\ S_5 & L_{33} \end{bmatrix} \prec 0, \quad \begin{bmatrix} L_{13} & S_6^T \\ S_6 & L_{31} \end{bmatrix} \prec 0, \quad 3L_{22} \preceq S_5^T + S_5 + S_6^T + S_6 \quad (15e)$$

Then the following LMI holds for $(\alpha, \beta) \in [0, 1]^2$:

$$\begin{aligned} L(\alpha, \beta) = & \quad \beta^2 \alpha^2 L_{11} + \beta^2 \alpha (1 - \alpha) L_{12} + \beta^2 (1 - \alpha)^2 L_{13} \\ & + \beta (1 - \beta) \alpha^2 L_{21} + \beta (1 - \beta) \alpha (1 - \alpha) L_{22} + \beta (1 - \beta) (1 - \alpha)^2 L_{23} \\ & + (1 - \beta)^2 \alpha^2 L_{31} + (1 - \beta)^2 \alpha (1 - \alpha) L_{32} + (1 - \beta)^2 (1 - \alpha)^2 L_{33} \prec 0. \end{aligned}$$

Proof: Applying Lemma 1 repeatedly to (15a) to (15d) then gives the following LMIs holds for all $\alpha \in [0, 1]$ and $\beta \in [0, 1]$:

$$\alpha^2 L_{11} + 3\alpha(1 - \alpha)L_{12} + (1 - \alpha)^2 L_{13} \prec 0, \quad (16a)$$

$$\alpha^2 L_{31} + 3\alpha(1 - \alpha)L_{32} + (1 - \alpha)^2 L_{33} \prec 0, \quad (16b)$$

$$\beta^2 L_{11} + 3\beta(1 - \beta)L_{21} + (1 - \beta)^2 L_{31} \prec 0, \quad (16c)$$

$$\beta^2 L_{13} + 3\beta(1 - \beta)L_{23} + (1 - \beta)^2 L_{33} \prec 0. \quad (16d)$$

Pre-and-post multiplying the first LMI in (15e) by $\begin{bmatrix} \frac{\alpha\beta}{(1-\alpha)(1-\beta)} I & \frac{(1-\alpha)(1-\beta)}{\alpha\beta} I \end{bmatrix}$ and its transpose:

$$\begin{aligned} & \frac{\alpha\beta}{(1-\alpha)(1-\beta)} L_{11} + \frac{(1-\alpha)(1-\beta)}{\alpha\beta} L_{33} + S_5^T + S_5 \prec 0, \\ & \alpha^2 \beta^2 L_{11} + (1-\alpha)^2 (1-\beta)^2 L_{33} + \alpha(1-\alpha)\beta(1-\beta)(S_5^T + S_5) \prec 0. \end{aligned}$$

Similar procedure applies to the second LMI in (15e) and together with the last LMI in (15e) gives:

$$\alpha^2 \beta^2 L_{11} + (1-\alpha)^2 \beta^2 L_{13} + \alpha^2 (1-\beta)^2 L_{31} + (1-\alpha)^2 (1-\beta)^2 L_{33} + 3\alpha(1-\alpha)\beta(1-\beta)(S_5^T + S_5) \prec 0. \quad (17)$$

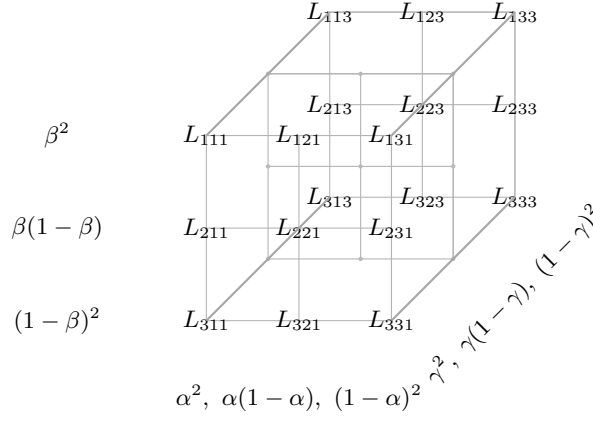
$\beta^2 \times (16a) + (1-\beta)^2 \times (16b) + \alpha^2 \times (16c) + (1-\alpha)^2 \times (16d) + (17)$ gives:

$$3L(\alpha, \beta) \prec 0.$$

■

Remark 2: The basic idea is, we know that $L_{11}, L_{13}, L_{31}, L_{33} \prec 0$, because they are the LMI conditions evaluated at the vertices of the set $[0, 1]^2$. But can we relax the other five points by not requiring them to be negative definite? The answer is yes, but it requires 6, i.e., $\binom{4}{2}$, slack matrices $S_1, \dots, S_6 \in \mathbb{R}^{n \times n}$ LMIs. Furthermore, in (15), the number 3 appears in the right-hand side of the inequalities instead of 1. This is because $L_{11}, L_{13}, L_{31}, L_{33}$ are counted multiple times in the combination, and for symmetric reason, the number is chosen to be uniform: $2^2 - 1 = 3$.

For the three-parameter case, a cube-like arrangement of L_{ijk} can be illustrated as follows.



The higher dimensional case follows as the hypercube-like arrangement of $L_{i_1 i_2 \dots i_\ell}$, which go over all possible combinations of pivots of the cube. For ℓ -parameter case, the number of pivots that are required to be negative definite is 2^ℓ , and the number of slack matrices is $\binom{2^\ell}{2}$ (all unordered vertex pairs), and the number in the right-hand side of the inequalities is $2^\ell - 1$. So the fundamental idea becomes:

Traverse all the pairs of the pivots of the hypercube, use slack matrices to relax the condition of the non-pivot points, and then combine all the LMIs together to get the final result.

Let us introduce the binary index function to facilitate the notation that runs over all the pivots of the hypercube:

Definition 1: For a ℓ -dimensional vector $\mathbf{i}$, the binary index function is defined as:

$$\text{ind}([\mathbf{i}]) = \sum_{k=1}^{\ell} \frac{i_k - 1}{2} 2^{\ell-k}, \quad \mathbf{i} = [i_1, \dots, i_\ell] \in \{1, 3\}^\ell.$$

Theorem 2: Suppose there exists $S_{[\mathbf{i}, \mathbf{j}]}$ such that:

$$\begin{bmatrix} L_{[\mathbf{i}]} & S_{[\mathbf{i}, \mathbf{j}]}^T \\ S_{[\mathbf{i}, \mathbf{j}]} & L_{[\mathbf{j}]} \end{bmatrix} \prec 0, \quad \forall \mathbf{i}, \mathbf{j} \in \{1, 3\}^\ell, \text{ind}([\mathbf{i}]) < \text{ind}([\mathbf{j}]), \quad (18a)$$

$$(2^\ell - 1)L_{[\mathbf{k}]} \preceq \sum_{\substack{\mathbf{i} + \mathbf{j} = 2\mathbf{k} \\ \mathbf{i}, \mathbf{j} \in \{1, 3\}^\ell \\ \text{ind}([\mathbf{i}]) < \text{ind}([\mathbf{j}])}} S_{[\mathbf{i}, \mathbf{j}]}^T + S_{[\mathbf{i}, \mathbf{j}]}, \quad \forall \mathbf{k} \in \{1, 2, 3\}^\ell \setminus \{1, 3\}^\ell, \quad (18b)$$

Then

$$L(\alpha_1, \dots, \alpha_\ell) = \sum_{i_1=1}^3 \dots \sum_{i_\ell=1}^3 \left[\prod_{j=1}^{\ell} \alpha_j^{3-i_j} (1 - \alpha_j)^{i_j-1} \right] L_{[i_1, i_2, \dots, i_\ell]} \prec 0, \quad \forall (\alpha_1, \dots, \alpha_\ell) \in [0, 1]^\ell.$$

Remark 3: The total dimension of the LMIs in (18) grows at

$$\underbrace{\frac{2^\ell(2^\ell - 1)}{2}}_{\text{\# of pairs in (18a)}} 2n + \underbrace{(3^\ell - 2^\ell)}_{\text{\# of non-pivot points in (18b)}} n = 2^{\ell+1} (2^{\ell-1} - 1) n + 3^\ell n.$$

For comparison, a brutal way is to treat $S_{[i_1, \dots, i_\ell]} \equiv 0$, which corresponds to the dimension of the LMI grows at

$$n \times 3^\ell.$$

So basically, the proposed method trades off for more number of LMIs with the conversation of brutally setting all $L_{[i_1, \dots, i_\ell]} \prec 0$. In reality, we do not need to consider all non-pivot points. For example, some parameter ρ_{i_k} does not change too much, in that case, one can consider setting

$$L_{[i_1, \dots, i_k=1, 2, 3, \dots, i_\ell]} \prec 0, \quad \forall i_1, \dots, i_{k-1}, i_{k+1}, \dots, i_\ell \in \{1, 2, 3\},$$

which reduce # of ℓ to $\ell - 1$.

2 Software idea

Based on the development above, I am thinking about write a generic software package that implements the gridding-based approximation for DP-LMIs, maybe starting from the scalar-parameter case first.

I searched on Internet for solvers that solves LPV problem. Turns out that there exist two different kinds: polynomial-based toolboxes (e.g., ROLMIP) and LFT-based toolboxes (e.g., LPVTools, R-RoMulOC).

- Polynomial-based toolboxes typically assume that all matrices and decision variables are represented in a polynomial basis:

$$P(\rho) = \sum_i a_i P_i, \sum_i a_i = 1. \quad (19)$$

By collecting and comparing the monomials, the original DP-LMI can be converted into a set of polynomial parameter-dependent LMIs. When there is only two a_i s, it is equivalent to the gridding-based approach with only one sub-interval $K = 1$.

- LFT-based toolboxes typically assume that the LPV system is represented in a Linear Fractional Transformation (LFT) form. The solvers relax the DP-LMI using a technique called ‘Integral Quadratic Constraints’(IQC). The issue with this approach is that the IQC-based relaxation is often very conservative.

Both routes are different modeling from the interpolation/gridding approach adopted in this report. In particular, for a typical LPV system, the polynomial-based approach will require a high-order polynomial basis to capture the parameter dependence, while the gridding-based approach can achieve a good approximation with a relatively coarse grid. On the other hand, the LFT-based approach may not be able to capture the parameter dependence accurately, leading to conservative results.

2.1 Scope and Core Features

The first version of the package is intentionally focused on solving a general problem in (3) with $\ell = 1$. The main components are:

1. Grid construction on Θ (scalar and multivariate) and automatic generation of interpolation coefficients.
2. Automatic computation of residual bounds r_0^k, r_1^k, r_2^k on each grid cell.
3. Direct linear-spline approximation for differential terms, consistent with the formulation in (3).
4. Automatic assembly of gridding LMIs (vertex and non-vertex constraints), including the multivariate extension.
5. Optional relaxation mode using the proposed lemma family (from the 1D case to the multivariate extension), to reduce conservatism compared with brute-force non-pivot negativity constraints.
6. Solver backend interface (e.g., YALMIP + SDP solvers), while keeping the front-end problem definition aligned with the gridding workflow.

2.2 A Comparison Example

Take the example from Masubuchi’s paper[5]:

$$\left[\begin{array}{c|c} A(\rho) & B(\rho) \\ \hline C(\rho) & D(\rho) \end{array} \right] = \left[\begin{array}{cc|cc} -1 & 0.5 & 1 & -4 \\ -1 & -2 & -1 & -1 \\ \hline 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{array} \right] + \rho \left[\begin{array}{cc|cc} -1.3 & -20 & 2.2 & 0.5 \\ 2 & -10 & -6 & -5 \\ \hline 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

The related DP-LMI is given by:

$$\left[\begin{array}{ccc} A(\rho)^T P(\rho) + P(\rho) A(\rho) + \varphi \frac{\partial P(\rho)}{\partial \rho} & P(\rho) B(\rho) & C^T(\rho) \\ B(\rho)^T P(\rho) & -\gamma I & D^T(\rho) \\ C(\rho) & D(\rho) & -\gamma I \end{array} \right] \prec 0, \forall \rho \in [0, 1], \varphi = \{-1, 1\}$$

I am going to compare 5 different methods:

1. **ROLMIP (polynomial baseline)**. Use **ROLMIP** as the polynomial-basis baseline. ROLMIP is designed for optimization problems with parameter-dependent variables over the unit simplex, where the variables are represented by homogeneous polynomials and the constraints are written as LMIs. In this package, you can expect the solution $P(\rho)$ to be represented as a homogeneous polynomial matrix as in (19).

2. **ROLMIP + Pólya's relaxation.** Use the same ROLMIP polynomial formulation. When $P(\rho)$ in (19) only admits two α_i terms, it is equivalent to the gridding-based approach with only one sub-interval $K = 1$, in which you can expect the γ overlaps.

In homogeneous polynomial, instead of relaxing the requirement of $L_2 < 0$ as we did in (13), another relaxation arises from positive polynomial is by Pólya's relaxation. For a homogeneous polynomial matrix $P(\alpha)$, the relaxation can be written informally as

$$P(\alpha) \succ 0, \quad \alpha \in \Lambda_m \quad \Leftarrow \quad (\alpha_1 + \cdots + \alpha_m)^d P(\alpha) \text{ has positive-definite coefficient matrices.}$$

Here d is the relaxation degree. Increasing d gives a less conservative certificate over the simplex, but it also introduces more monomial coefficient constraints, which grows combinatorially with d and m . In this example, we will use $d = 2$.

3. **Gridding (proposed interpolation route).** Apply the gridding/spline approximation developed in this report. The DP-LMI is replaced by the finite set of grid-based LMIs in (4), so this is the direct implementation of the proposed interpolation route.
4. **Gridding + relaxation (proposed relaxed version).** Start from the gridding formulation above, but use the slack-matrix relaxation in (13). This keeps the same interpolation structure while reducing the conservatism caused by requiring all non-pivot grid conditions to be negative definite, especially when the grid is coarse.
5. **LPVTools benchmark (LFT/IQC route).** Model the same plant in LPVTools. LPVTools is a MATLAB toolbox for LPV modeling, model reduction, analysis, synthesis, and simulation. This benchmark represents a mature LPV toolbox workflow whose modeling philosophy is different from both the polynomial ROLMIP route and the gridding/interpolation route proposed here. It heavily relies on the LFT/IQC relaxation technique.

The result of comparison is given in Fig 1. Both gridding technique and ROLMIP starts at the same $\gamma = 7.58491073869303$ because when polynomial degree/gridding segment number is 0, it represents constant $P(\rho) \equiv P$. When polynomial degree/gridding segment number is 1, ROLMIP and gridding technique are equivalent, so they both give $\gamma = 6.79910188692992$. You would notice that both relaxation (Pólya's for ROLMIP and slack-matrix for gridding) reduces the conservatism. Overall, the gridding-based approach with relaxation achieves the best performance among the five methods, demonstrating the effectiveness of the proposed interpolation and relaxation techniques in handling DP-LMIs for LPV systems.

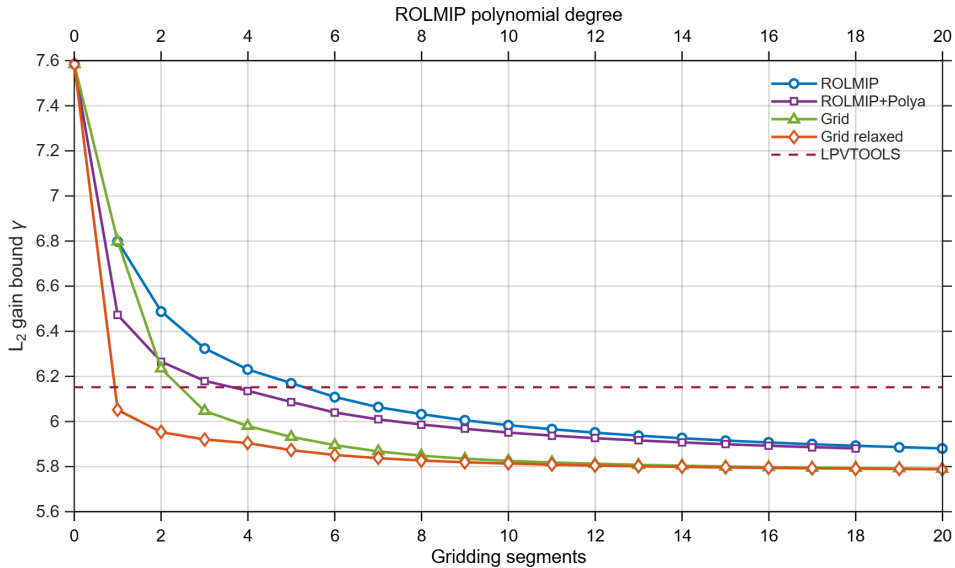


Figure 1: Comparison of different methods

The relaxation of Pólya's method is more expensive which broke the solver when the polynomial degree is larger than 18, which results to the missing of the purple line in Fig 1 after degree 18.

To see the difference between the homogeneous polynomial approach and the gridding-based approach, we can directly see the structure of $P(\rho)$ of degree 2 that is obtained from the two methods Fig 2 where the gridding-based approach with relaxation exhibits a more flexible structure compared with the homogeneous polynomial approach with poly's relaxation, which is constrained by the fixed monomial basis.

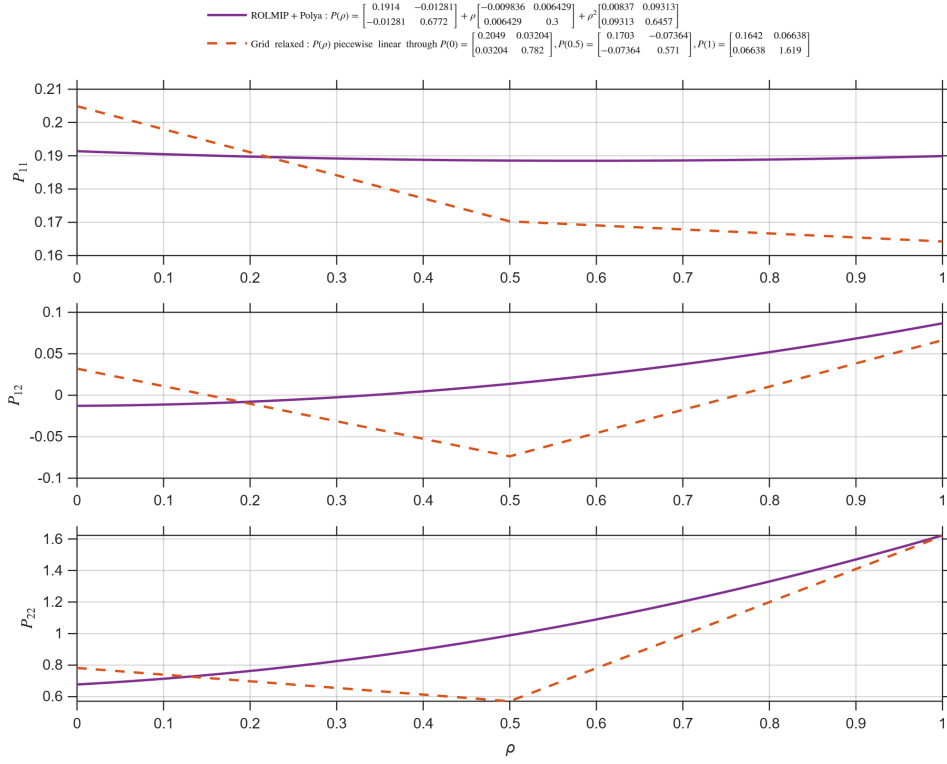


Figure 2: Comparison of $P(\rho)$ obtained from different methods

References

- [1] Solmaz Sajjadi-Kia and Faryar Jabbari. Controllers for linear systems with bounded actuators: Slab scheduling and anti-windup. *Automatica*, 49(3):762–769, March 2013.
- [2] Daniel Liberzon. *Switching in Systems and Control*. Birkhäuser Boston, Boston, MA, 2003.
- [3] M.C. de Oliveira, J. Bernussou, and J.C. Geromel. A new discrete-time robust stability condition. *Systems & Control Letters*, 37(4):261–265, July 1999.
- [4] Yicheng Xu and Faryar Jabbari. Spline-based parameter varying output feedback synthesis with improved L_2 gain. In *2024 IEEE 63rd Conference on Decision and Control (CDC)*, pages 1455–1460. IEEE, December 2024.
- [5] I. Masubuchi, A. Kume, and E. Shimemura. Spline-type solution to parameter-dependent LMIs. In *Proceedings of the 37th IEEE Conference on Decision and Control (Cat. No.98CH36171)*, volume 2, pages 1753–1758. IEEE, 1998.